

Integer Arithmetic and Performance Basics

Addition, Subtraction, Multiplication, Division, Shifts, NZCV Flags, and Execution Time Metrics in AArch64

Topic 4 Outline

Topic Objective

Analysis of how arithmetic operations manipulate fixed-width bit patterns, how condition flags govern control flow, and how hardware performance is measured using instruction counts, CPI, and clock rates.

- **Section 1:** Integer Addition, Subtraction, and Unsigned Wraparound
- **Section 2:** Two's Complement Arithmetic and Signed Overflow
- **Section 3:** AArch64 Condition Flags (NZCV) and Flag-Setting Instructions
- **Section 4:** Multiplication, Division, and Logical/Arithmetic Shifts
- **Section 5:** Processor Performance: The Iron Law and Instruction Mix
- **Section 6:** Summary and Worked Self-Test Exercises

Section 01

Integer Addition, Subtraction, and Unsigned Wraparound

Why Integer Arithmetic Matters

Integer arithmetic executes continuously during program execution:

- **Core Operations:** Memory addresses, loop counters, array indexing, and comparisons rely entirely on integer manipulation.
- **Fixed-Width Registers:** AArch64 registers hold fixed-width values (32-bit w or 64-bit x).
- **Arithmetic Effects:** Operations can wrap around, generate carries, or trigger signed overflows.
- **Control Flow Integration:** Condition flags connect arithmetic outputs directly to subsequent control decisions.

Binary Addition and Unsigned Ranges

Addition Mechanics

- Proceeds from least significant bit (LSB) to most significant bit (MSB).
- $1 + 1$ produces 0 with a carry out of 1.
- Carries propagate across higher-order bit positions.

Unsigned Addition & Carry Out

- Treats all bits strictly as magnitude.
- An n -bit destination keeps only the low-order n bits.
- A carry-out indicates the mathematical result exceeded the range 0 to $2^n - 1$.

Example (32-Bit Unsigned)

$$0xFFFFFFFF + 0x00000001 = 0x00000000 \text{ with } C = 1$$

Binary Subtraction Using Two's Complement

Hardware avoids dedicated subtraction logic by reusing addition circuitry:

- Subtraction ($A - B$) is computed as $A + (-B)$.
- To form $-B$, invert all bits of B (one's complement) and add 1.
- Any carry generated beyond the register width is discarded from the stored container.

4-Bit Subtraction Example ($6 - 3$)

- $+3 = 0011_2 \rightarrow -3 = \text{invert}(0011) + 1 = 1100 + 1 = 1101_2$.
- Compute $6 + (-3)$: $0110_2 + 1101_2 = 1\ 0011_2$.
- Discarding the carry bit yields stored result $0011_2 = +3_{10}$.

Two's Complement Arithmetic and Signed Overflow

Signed vs. Unsigned Interpretation

The ALU operates exclusively on raw bit patterns without inherent data types:

- The exact same bit container can be interpreted as either signed or unsigned.
- Signed and unsigned addition execute via the identical underlying binary adder.
- Carry and overflow flags interpret the output differently depending on intended data type.

Example (8-Bit Pattern: $1111\,1111_2$)

- Unsigned interpretation equals 255.
- Signed two's complement interpretation equals -1 .

Detecting Signed Overflow

Definition of Signed Overflow

Signed overflow occurs when the true mathematical result of a two's complement addition falls outside the representable signed range.

- **Rule of Thumb:** Adding two positive values must not produce a negative result; adding two negative values must not produce a positive result.
- Hardware records signed overflow in the **V** (Overflow) flag.

8-Bit Overflow Example

$$0111\ 1111_2(+127) + 0000\ 0001_2(+1) = 1000\ 0000_2(-128), \text{ with } V = 1$$

Carry versus Signed Overflow

Carry and overflow evaluate distinct numerical conditions and are not interchangeable:

Carry Flag (C)

- Relates primarily to unsigned arithmetic.
- Records carry out of the most significant bit.
- Example: $0xFF + 0x01 = 0x00$, with $C = 1$ (unsigned wrap).

Overflow Flag (V)

- Relates to signed two's complement arithmetic.
- Indicates a sign violation exceeding container bounds.
- Example: $0x7F + 0x01 = 0x80$, with $V = 1$ (signed overflow).

AArch64 Condition Flags (NZCV) and Flag-Setting Instructions

Processor State and Condition Flags (NZCV)

Arithmetic results automatically update condition flags stored in **PSTATE**:

Flag	Condition Evaluated	Technical Meaning
N (Negative)	Result is negative signed value	Copied from MSB (bit 63 or 31)
Z (Zero)	Result equals exactly zero	Set (1) if all result bits are zero
C (Carry)	Unsigned carry or borrow	Unsigned overflow / carry out from MSB
V (Overflow)	Signed two's complement overflow	Signed range violation detected

Flag Persistence

Once set, the NZCV flag pattern persists across subsequent instructions until modified by another arithmetic or comparison instruction.

Flag-Setting Instructions: `adds`, `subs`, and `cmp`

Standard instructions compute results without modifying flags unless explicitly requested:

The `s` Suffix Variant

- `adds x0, x1, x2`: Performs addition, stores sum in `x0`, and updates **NZCV** flags.
- `subs x3, x3, #1`: Subtracts and updates flags.

Comparison (`cmp`)

- `cmp x4, x5` is an alias for `subs xzr, x4, x5`.
- Subtracts operands to set flags, but **discards the result**.
- Feeds conditional branches like `b.eq` or `b.lt`.

Section 04

Multiplication, Division, and Shifts

Binary Multiplication Mechanics

- Multiplication operates as repeated shifted additions.
- Each multiplier bit selects a shifted copy of the multiplicand.
- Multiplying two n -bit values produces a full product up to $2n$ bits wide.
- Fixed-width registers typically retain only the lower n bits of the product.

Example (5×3)

$$\begin{aligned} &0101_2 \times 0011_2 \\ &= 0101_2 + 1010_2 \text{ (shifted)} \\ &= 1111_2 = 15_{10} \end{aligned}$$

AArch64 Multiply and Divide Instructions

AArch64 provides dedicated hardware instructions for arithmetic expansion:

Instruction Syntax	Operation Performed	Architectural Effect
<code>mul x0, x1, x2</code>	64-Bit Multiplication	$x0 \leftarrow x1 \times x2$
<code>udiv x3, x4, x5</code>	Unsigned Division	$x3 \leftarrow x4/x5$ (unsigned)
<code>sdiv x3, x4, x5</code>	Signed Division	$x3 \leftarrow x4/x5$ (signed)

Microarchitectural Note

Multiplication and division instructions typically exhibit higher instruction latency than simple single-cycle additions.

Logical and Arithmetic Shift Operations

Shifts move bit patterns within a register to execute rapid scaling or isolation:

Logical Shift Left (lsl)

- Shifts bits left.
- Fills vacant low-order bits with zeros.
- Multiplies by powers of two (2^n).

Logical Shift Right (lsr)

- Shifts bits right.
- Fills vacant high-order bits with zeros.
- Unsigned division by 2^n .

Arithmetic Shift Right (asr)

- Shifts bits right.
- Preserves sign bit in high-order positions.
- Signed division by 2^n .

Processor Performance: The Iron Law and Instruction Mix

Processor Performance Basics

The Ultimate Metric: Execution Time

Performance is measured strictly by total program execution time, not by clock frequency alone.

- Processors execute ordered sequences of machine instructions.
- Each instruction takes one or more clock cycles depending on complexity and hazards.
- **CPI (Cycles Per Instruction)** summarizes the average execution cycles required per instruction across a workload.
- Instruction count, CPI, and clock rate must be considered together.

The Iron Law of Processor Performance

Deconstructing CPU Execution Time

Performance is governed by the product of three fundamental metrics:

$$\text{CPU Time} = \frac{\text{Instructions}}{\text{Program}} \times \frac{\text{Clock Cycles}}{\text{Instruction (CPI)}} \times \frac{\text{Seconds}}{\text{Clock Cycle}}$$

- **Instruction Count:** Determined by ISA efficiency, compiler optimization level, and source algorithm complexity.
- **Average CPI:** Determined by microarchitecture design (pipelining, memory hierarchy, functional unit latencies).
- **Clock Cycle Time:** Determined by semiconductor technology, circuit design, and critical path propagation delays.
- **Trade-off Caution:** Optimizing one variable often degrades another (e.g., aggressive compiler optimizations might lower instruction count but increase average CPI).

Advanced Analysis: Variable CPI Workloads

When instructions have differing latencies, average CPI is calculated via the instruction mix:

$$\text{Average CPI} = \sum_{i=1}^n (\text{CPI}_i \times \text{Fraction}_i)$$

Challenging Example

A program executes 10^9 instructions with three classes:

- **Class A** (CPI = 1): 50% of instructions
- **Class B** (CPI = 2): 30% of instructions
- **Class C** (CPI = 4): 20% of instructions

Calculate Average CPI & Execution Time (2 GHz Clock):

- Avg CPI = $(1 \times 0.5) + (2 \times 0.3) + (4 \times 0.2) = 0.5 + 0.6 + 0.8 = 1.9$
- CPU Time = $\frac{10^9 \times 1.9}{2 \times 10^9 \text{ cycles/s}} = 0.95 \text{ seconds}$

Summary & Self-Test

Topic 4 Comprehensive Summary

- **Binary Addition & Subtraction:** Standard binary addition handles unsigned logic, while two's complement encoding allows the same adder to execute subtraction ($A + (-B)$).
- **Signed Overflow:** Occurs when results exceed container bounds, tracked independently from unsigned carries via the V flag.
- **Condition Flags:** Instructions with the s suffix and `cmp` update **NZCV** flags in `PSTATE` to drive conditional branching.
- **Arithmetic Extensions:** Multiplication (`mul`), division (`udiv/sdiv`), and shifts (`lsl/lshr/asr`) expand ALU capabilities.
- **Iron Law Performance:** CPU execution time is fundamentally tied to Instruction Count, average CPI (via instruction mix), and Clock Rate.

Self-Test: Questions 1 & 2

Question 1: Addition and Carry

How does binary addition produce a carry, and what does the carry-out indicate for unsigned integer arithmetic?

Question 2: Two's Complement Subtraction

How can subtraction be performed using two's complement addition without requiring dedicated hardware subtraction circuitry?

Self-Test: Solutions 1 & 2

Solution 1

Binary addition generates a carry when the sum of bits in a column equals or exceeds 2, propagating a 1 into the next higher-order column. For unsigned arithmetic, a final carry-out indicates the result exceeded the representable range 0 to $2^n - 1$.

Solution 2

Subtraction ($A - B$) is computed as $A + (-B)$. The negative operand $-B$ is formed by inverting all bits of B (one's complement) and adding 1, allowing the standard binary adder to process subtractions directly.

Self-Test: Questions 3 & 4

Question 3: Carry vs. Signed Overflow

What is the fundamental difference between carry (C) and signed overflow (V)? Can an operation have carry without overflow?

Question 4: Flag-Setting Instructions

What is the difference between `add` and `adds`, and how does `cmp` function as an alias for flag generation?

Self-Test: Solutions 3 & 4

Solution 3

Carry tracks unsigned range overflow (carry out of MSB), whereas signed overflow (V) tracks two's complement sign violations. Yes, adding large unsigned numbers can set $C = 1$ without violating signed bounds ($V = 0$).

Solution 4

`adds` computes the arithmetic sum and updates the NZCV condition flags in PSTATE, whereas standard `add` does not update flags. `cmp` subtracts operands to update flags for branching but discards the numeric result.

Self-Test: Questions 5 & 6

Question 5: Shifts and Powers of Two

What is the difference between `lsl`, `lsr`, and `asr`? When is each appropriate for multiplying or dividing by powers of two?

Question 6: CPU Performance Calculation

Consider two processors running the same 1-million instruction program:

- Processor A: $\text{CPI} = 2$, Clock Rate = 2 GHz
- Processor B: $\text{CPI} = 1$, Clock Rate = 1 GHz

Which processor has the lower execution time?

Self-Test: Solutions 5 & 6

Solution 5

lsl shifts left (multiplies by 2^n), lsr shifts right filling zeros (unsigned division by 2^n), and asr shifts right preserving the sign bit (signed division by 2^n).

Solution 6

- Processor A Time: $(10^6 \times 2)/(2 \times 10^9) = 0.001 \text{ s (1 ms)}$.
- Processor B Time: $(10^6 \times 1)/(1 \times 10^9) = 0.001 \text{ s (1 ms)}$.
- Both processors achieve identical execution time.

Wrap-Up and Next Steps

Transition to Topic 5

Building upon integer arithmetic, condition flags, and performance equations, the next module examines:

- **AArch64 Arithmetic and Logical Instructions:** Advanced exploration of ALU data manipulation mechanics.
- **Arithmetic Instructions:** Executing signed and unsigned addition, subtraction, and extended math operations.
- **Logical and Bitwise Instructions:** Performing bitwise operations, masking, and boolean evaluations.
- **Shift Instructions:** Manipulating bit positions using logical shifts, arithmetic shifts, and rotates.
- **Immediate Values and Constants:** Encoding constants directly within instructions and utilizing wide constant loading sequences.